

Sun Woo (P.) Kim

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Education

PhD in Physics, [King's College London](#) (visiting [University of Cambridge](#)) 2023-

Many-body physics: statistical mechanics, quantum information, machine learning.

Working with [Curt von Keyserlingk](#) of King's College London and [Austen Lamacraft](#) of University of Cambridge

MASt in Physics, [University of Cambridge](#) 2018-2019

Distinction.

Notable courses: Theories of Quantum Matter, Quantum Field Theory, Quantum Information

BSc Physics with Theoretical Physics, [Imperial College London](#) 2015-2018

1st Class (80.7%), Dean's List for all three years.

Notable courses: Foundations of Quantum Mechanics, General Relativity, Complexity and Networks

International Baccalaureate, [United World College of South East Asia Dover Campus](#) 2012-2014

41/45 (91%). Additional Standard Chemistry 6/7.

7 Subjects, Higher Physics 7/7, Higher Mathematics 7/7, Higher Geography 7/7, Standard English 6/7.

Employment/Experience

Research scientist at [AIRS Medical](#) (Republic of Korea national service) 2019-2023

- Part of National service in Republic of Korea as a 'Skilled Industry Personnel', applying machine learning to medical imaging and diagnostic settings.
- Came 1st for all tracks in the 2020 Facebook FastMRI Challenge, see publications for details. Published [7 patents](#).

Research student at [Max Planck Institute of Complex Systems](#) 2019-2022

- Worked with [Markus Heyl](#) and Giuseppe De Tomasi in many-body localization, see publications for details.
- This research project was done part-time during my time in Republic of Korea's national service.

Undergraduate research project: group theoretic analysis of structured elastic plates 2018

- Worked with [Richard Craster](#), Mehul Makwana. Research in metamaterials using representation theory and topology.
- Awarded the UROP Prize in Mathematics.

Publications

S.W.P. Kim and M. McGinley, [Rigorous mixed-state topological order beyond Pauli stabiliser codes and Pauli noise channels](#), *In. prep.* (2026)

F. Huebner and S.W.P. Kim, [A systematic theory of influence solvability and its application to brickwork circuits](#), *In. prep.* (2026)

S.W.P. Kim, [Optimal recovery for quantum error correction](#), *arXiv preprint arXiv:2603.06520* (2026)

S.W.P. Kim and G. Pinna, [Existence and bounds of growth constants for restricted walks, surfaces, and generalisations](#), *arXiv preprint arXiv:2509.04568* (2025)

S.W.P. Kim, A. Lamacraft, and C. von Keyserlingk, [Measurement-induced phase transitions in quantum inference problems and quantum hidden Markov models](#), *arXiv preprint arXiv:2504.08888* (2025)

S.W.P. Kim, F. Huebner, J.P. Garrahan and B. Doyon, [Circuits as a simple platform for the emergence of hydrodynamics in deterministic chaotic many-body systems](#), *arXiv preprint arXiv:2503.08788* (2025)

S.W.P. Kim and A. Lamacraft, [Planted directed polymer: Inferring a random walk from noisy images](#), *Physical Review E*, 111 (2), 024135 (2025)

K. Tazi, S.W.P. Kim, M. Girona-Mata and R. E. Turner, [Refined climatologies of future precipitation over High Mountain Asia using probabilistic ensemble learning](#), *Environmental Research Letters* (2025)

S.W. Kim, G. De Tomasi and M. Heyl, [Real-time dynamics of one-dimensional and two-dimensional bosonic quantum matter deep in the many-body localized phase](#), *Physical Review B* 104 (14), 144205 (2021)

M.J. Muckley, B. Riemenschneider, A. Radmanesh, S. Kim et. al., [Results of the 2020 fastMRI Challenge for Machine Learning MR Image Reconstruction](#), *IEEE transactions on medical imaging*, 40(9), 2306–2317 (2021)

Awards

EPSRC DTP International Studentship	2023
Granted to one international applicant in the department at King's College London. Grant Ref no: EP/W524475/1	
E. M. Burnett Prize	2019
In recognition for obtaining Distinction in Master of Advanced Studies.	
UROP Prize in Mathematics	2018
Awarded to students of outstanding performance in the Undergraduate Research Opportunity Programme (UROP), for the project 'Group Theoretic Analysis of Structured Elastic Plates'.	
Dean's List for 1st, 2nd and 3rd Years	2016, 2017, 2018
Awarded for being the top 10% of students in cohort of 2017/18 of the Physics programme at Imperial College London.	

Talks

LPENS (Threefold Quantum: Theory, Experiment, Computation Out of Equilibrium workshop)	Mar 2026
A systematic theory of influence solvability and its application to brickwork circuits.	
UBC (Probability group seminar)	Sep 2025
Existence and bounds of growth constants for restricted walks, surfaces, and generalisations.	
Christ's College, Cambridge (Quantum computation through the lens of many-body physics workshop)	Sep 2025
Measurement-induced phase transitions in quantum inference problems.	
KCL (Probability group seminar)	Sep 2025
Existence and bounds of growth constants for restricted walks, surfaces, and generalisations.	
Budapest Integrability Seminar	Jul 2025
Circuits as a simple platform for the emergence of hydrodynamics in deterministic chaotic many-body systems.	
BME, Budapest (Tibor Rakovszky's group)	Jun 2025
Circuits as a simple platform for the emergence of hydrodynamics in deterministic chaotic many-body systems.	
University of Cambridge (CPGM at TCM)	May 2025
Circuits as a simple platform for the emergence of hydrodynamics in deterministic chaotic many-body systems.	
Harvard University (Norman Yao's group)	Apr 2025
Measurement-induced phase transitions in quantum inference problems.	
NYU (Dries Sel's group)	Apr 2025
Measurement-induced phase transitions in quantum inference problems.	
Princeton University (Sarang Gopalakrishnan's group)	Apr 2025
Measurement-induced phase transitions in quantum inference problems.	
KCL (Disordered Systems seminar)	Apr 2025
Planted directed polymer: inferring a random walk from noisy images.	
University of Cambridge (TCM Journal Club)	Dec 2024
Topological quantum memory and quantum error correction as a Bayesian inference problem.	
Institute of Physics (Advances in Quantum Transport in Low Dimensional Systems workshop)	Sept 2024
Measurement-induced phase transitions in quantum inference problems.	
SNU (Yongjoo Baek's Group)	May 2024
Planted directed polymer: inferring a random walk from noisy images.	
KIAS (Statistical physics seminar)	May 2024
Planted directed polymer: inferring a random walk from noisy images.	
Imperial College London (Theory of Topological Matter group seminar with Frank Schindler's group)	Apr 2024
Bayesian inference and statistical physics.	
KCL (Many Body Circle seminar)	Mar 2024
Polarons.	

Workshop: Medical Imaging Meets NeurIPS (NeurIPS 2020)

Dec 2020

Talk by best performing team for the 2020 FastMRI Challenge for Machine Learning MR Image Reconstruction

Other Experiences

Expert advisor for white paper by Tony Blair Institute for Global Change

2025

- White paper title: "[A New National Purpose: A UK Quantum Strategy for Sovereignty and Scale](#)"
- Alongside Toby Cubitt of Phasecraft, Jason Crain of IBM, and Gerald Mullally of OQC, among others.

Organiser for Many Body Circle Seminar Series/Journal Club

2023-

- Collaborated PhD students from condensed matter theory and disordered systems groups at KCL. Invited external speakers (Imperial, UCL)
- Past events at sites.google.com/view/kclmanybodycircle/events

Teaching

2023-

- **Fall 2025 (KCL):** Authored problem set questions for Classical Physics, a 1st year course.
- **Spring 2025 (Cambridge):** TA for Computational Physics, a 3rd year (Part II) course.
- **Spring 2025 (KCL):** TA for Mathematical Methods for Theoretical Physics, a 2nd year course.
- **Fall 2024 (Cambridge):** TA for Theories of Quantum Matter, a master's (Part III) course.
- **Spring 2024 (KCL):** TA for Mathematical Methods for Theoretical Physics, a 2nd year course.
- **Spring 2024 (KCL):** TA for Symmetry in Physics, a 2nd year course.
- **Fall 2023 (KCL):** TA for Mathematical Methods for Physics, a 2nd year course at KCL. Worked through example questions in lecture theatre of 40+ students.

OUTREACH Mentoring Scheme

2016-2017

- Mentored students and prospective students on various areas such as Physics, Maths, and Computing. Worked with a group of mentors organising activities and demonstrations for 20 students.
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Skills

Computing

- Scientific programming. In python: NumPy, SciPy, Numba, and ML using PyTorch, PyTorch Lightning, TensorFlow, Stim in Python. In Julia: iTensor. Experience in MATLAB, Mathematica, Fortran, C++, git, LaTeX, Slurm.

Languages

- English (native fluency), Korean (native fluency)

Further Interests

Jazz Guitar

- Straight-ahead (bebop, hard-bop, latin), fusion, and contemporary. Played in small band (Duo, Trio, Quartet) and big band (Churchill Jazz Band, Jesus College Big Band) gigs. Attending jams in Seoul, London.

Map Designer for Starcraft II

2013-2014

- [Created official maps](#), such as Frost, Bridgehead, and Fruitland for real time strategy game, Starcraft II.
- Combined game knowledge with critical thinking to create effective, balanced, and dynamic maps that were used for over 4 years in the competitive scene, played in over 3000+ competitive matches.

Details are available upon request